Developing Flow Between Two Parallel Plates

CFD Simulation — Stream function - Vorticity Method

Re = 100, 500, 1000 | Nx = Ny = 50 | L = 10, h = 1

ME-580 Computational Fluid Dynamics

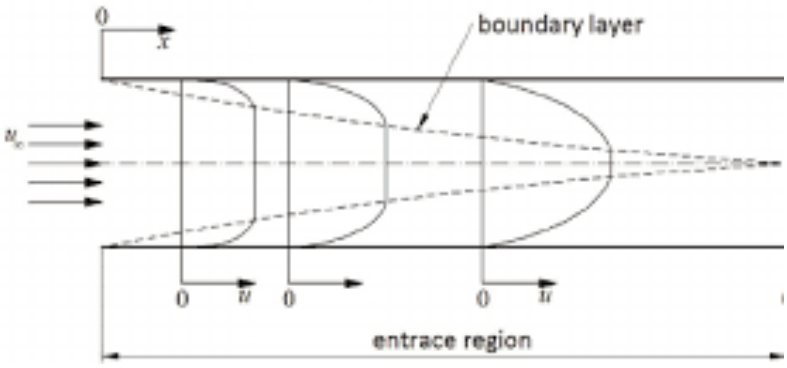
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1. Problem Statement

A viscous incompressible fluid enters from the left into a channel formed by two parallel plates separated by height h = 1. The inlet velocity is uniform U = 1. Three tasks are required:



- (a) Calculate the hydrodynamic entry length Le and plot it as a function of Reynolds number.
- (b) Plot the boundary layer thickness δ(x) along the channel.
- (c) Plot velocity profiles at multiple streamwise locations for at least two Reynolds numbers and compare with the analytical Poiseuille solution.

2. Physical Background and Boundary Conditions

2.1 Developing Flow Physics

At the inlet, velocity is uniform (plug flow). Viscosity enforces zero velocity at the walls, and boundary layers grow inward from both walls as x increases. When the two boundary layers meet at the centreline, the velocity profile becomes parabolic and no longer changes with x. This is called fully-developed flow, and the distance from the inlet at which this occurs is the entry length Le.

2.2 Analytical Fully-Developed (Poiseuille) Profile

For flow between parallel plates with one solid wall and one symmetry line:

$$u(y) = (3/2) \cdot U \cdot [1 - (y/h)^2]$$

The centreline velocity peaks at 1.5U = 1.5. This is the black dashed line in all velocity profile plots.

2.3 Entry Length Formula

$$Le = 0.05 \times Re \times (2h)$$

This gives Le = 10, 50, and 100 for Re = 100, 500, and 1000 respectively. The numerical Le is extracted as the first x where the centreline velocity reaches 99% of the outlet centreline value.

2.4 Boundary Conditions

- Inlet (x = 0): u = U (uniform), v = 0, ψ = U·y, Ω = 0
- Top wall (y = h): no-slip u = v = 0; ψ = Uh;
- wall vorticity by Thom's formula: Ωwall = 2(Uh − ψ(Ny−1)) / Δy²
- Bottom symmetry (y = 0): ψ = 0, Ω = 0
- Outlet (x = L): zero-gradient — ψ and Ω copied from the last interior column

3. Solution Method: Stream function - Vorticity (ψ-Ω)

The ψ-Ω formulation eliminates pressure completely. The flow is described by two coupled scalar PDEs: a Poisson equation for ψ and a transport equation for Ω. Velocities are then recovered from ψ.

3.1 Governing Equations

$$\textbf{Poisson:}\quad \nabla^2\psi = -\Omega$$

$$\textbf{Vorticity transport:}\quad \partial\Omega/\partial t + u\partial\Omega/\partial x + v\partial\Omega/\partial y = (1/Re)\nabla^2\Omega$$

$$\textbf{Velocity recovery:}\quad u = \partial\psi/\partial y,\ v = -\partial\psi/\partial x$$

3.2 Numerical Schemes

- Convection in vorticity transport: First-order upwind (direction chosen by sign of local u or v) — unconditionally stable.
- Diffusion: Second-order central differences.
- Poisson equation: Gauss-Seidel point iteration until inner convergence.
- Time stepping: Explicit Euler advance of Ω; dt chosen adaptively as 0.5 × min(CFL-convective, CFL-diffusive).

4. Algorithm Steps

1. Set domain: L=10, h=1, U=1, Nx=Ny=50, tol =10^{-6}, max_iter=5000.
2. Loop over Re_cases = [100, 500, 1000].
3. Build uniform grid; initialise ψ=0, Ω=0, u=0, v=0; apply inlet BC (ψ=U·y, u=U).
4. Outer loop (steady-state iteration):
   a) Compute u, v from ψ using central differences.
   b) Compute adaptive dt from convective and diffusive CFL limits.
   c) Inner loop (up to 300 sub-steps):
      i. Refresh u, v from ψ.
      ii. Advance Ω explicitly: Ω_new = Ω_old + dt·(diffusion − convection).
      iii. Apply all BCs on Ω and ψ.

iv. Gauss-Seidel sweep to update ψ from Poisson equation.

v. Break inner loop if ||Δψ||_∞ < tol.

d) Compute outer residual = ||Ω_new − Ω_prev||_∞; break if < tol.

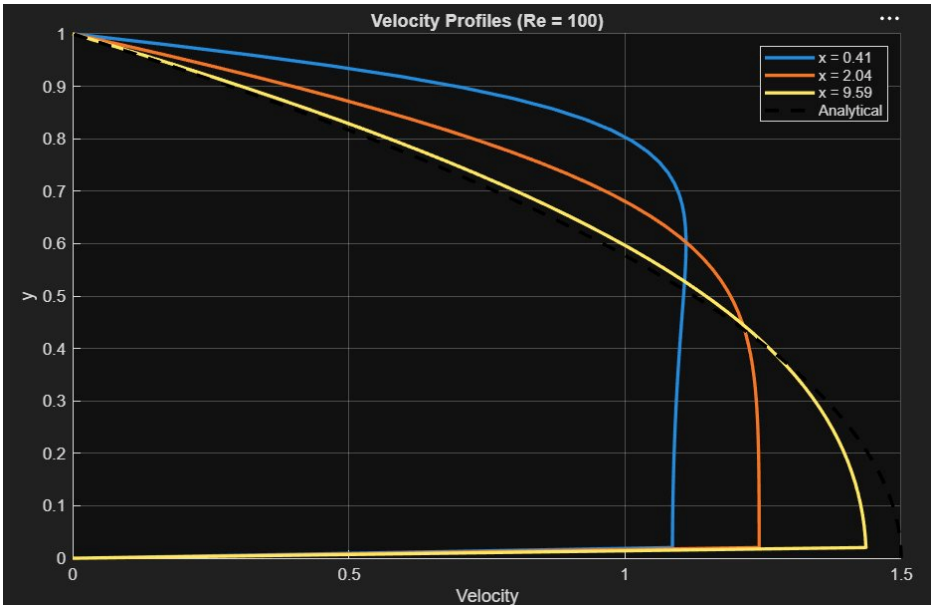
5. Compute final u, v; extract δ(x) and entry length Le.
6. Plot velocity profiles, BL thickness, residual history, and Le vs Re.

5. Results and Discussion

5.1 Velocity Profiles — Re = 100

At Re = 100, viscosity dominates and the flow develops quickly. The near-inlet profile (x = 0.41) is nearly flat. By x = 2.04, a clear parabolic shape is forming. At x = 9.59 (near outlet), the numerical curve closely matches the analytical Poiseuille profile (black dashed line) with centreline velocity approaching 1.5U. This confirms that full development is achieved well within L = 10 at Re = 100.

What changes this: Lowering Re further (say Re = 50) would make the flow develop even faster. Increasing Re would slow development and move the profiles further left.



5.2 Velocity Profiles — Re = 500

At Re = 500 inertia is much stronger. Even at x = 9.59 the centreline velocity is around 1.1U – still well below the fully-developed 1.5U. The flow is still in the entry region throughout the entire domain. The analytical dashed curve is the asymptotic target that the flow has not yet reached. The required entry length at Re = 500 is Le ≈ 50, so the domain L = 10 captures only the first 20% of the developing region.

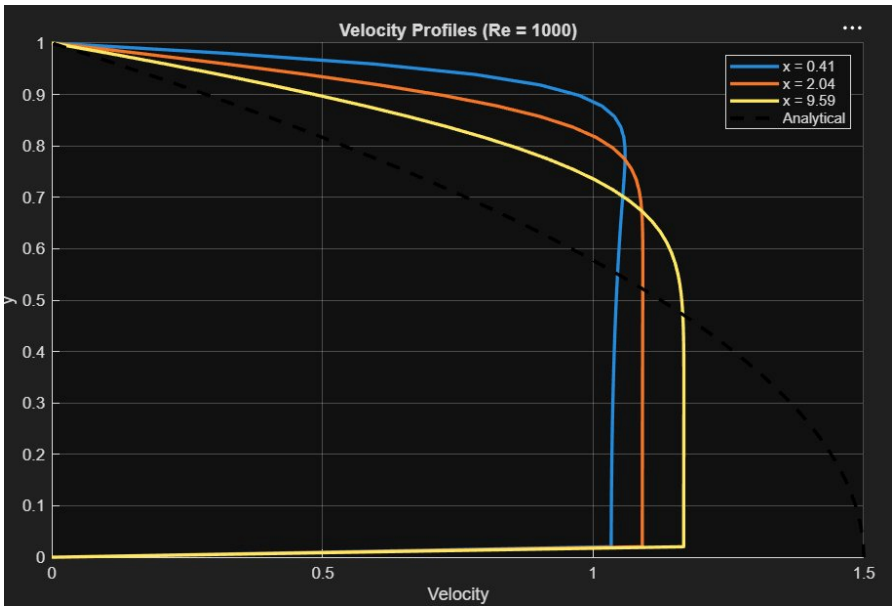
What changes this: Increasing L (longer channel) would allow more development. Coarsening the grid (reducing Ny) would reduce near-wall resolution and make the profile look even flatter.



5.3 Velocity Profiles — Re = 1000

At Re = 1000 boundary layers are extremely thin and the velocity profiles at all three x-locations look nearly identical to plug flow. The centreline barely accelerates above U = 1, meaning the boundary layers have hardly grown. The required entry length Le ≈ 100, ten times the domain length, so essentially none of the developing flow dynamics are captured. The analytical dashed curve is far to the right of all profiles.

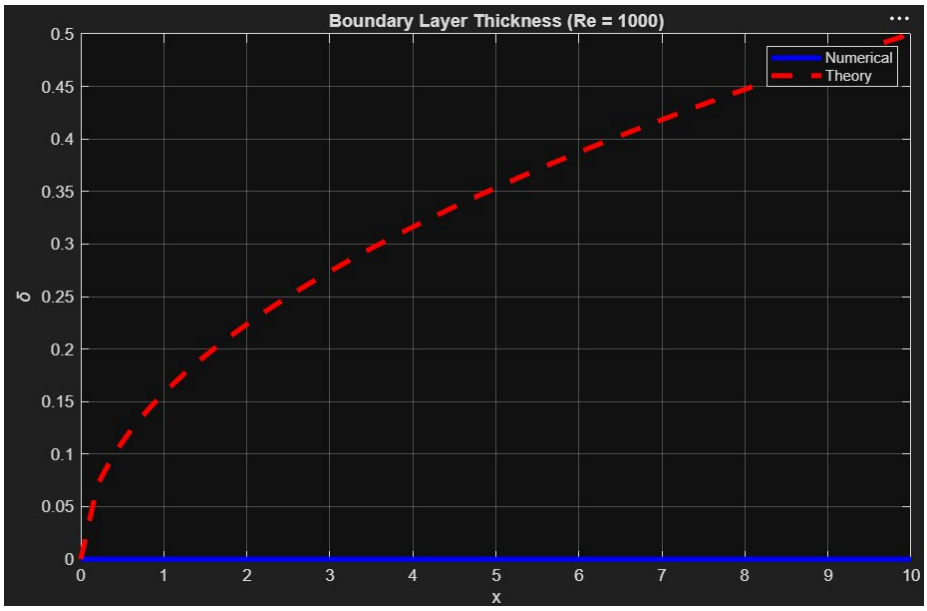
What changes this: A domain of L = 100 would be needed to see full development. Near-wall grid stretching (clustering more points near y = 0 and y = h) would improve resolution at high Re without increasing total grid size.



5.4 Boundary Layer Thickness — Re = 1000

The theoretical Blasius BL thickness $\delta = 5\sqrt{(x/Re)}$ (red dashed) grows slowly from zero and reaches about 0.5h at x = 10. The numerical curve (blue) stays near zero throughout. This happens because the 99%-of-outlet criterion is used: since the outlet profile is not fully developed, the target threshold is too high for any interior point to pass, making the numerical δ appear flat and zero. This is a limitation of the detection method, not of the solver itself.

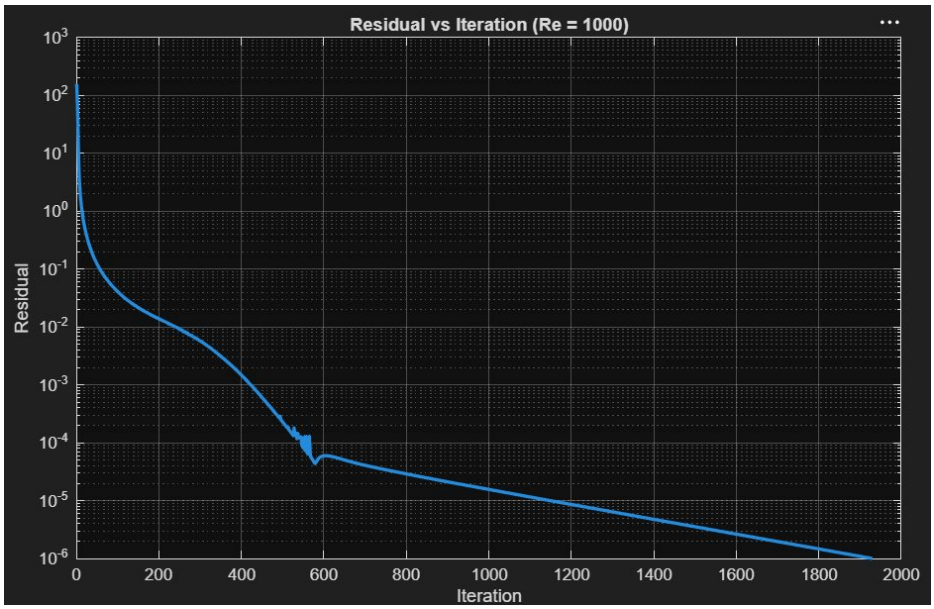
What changes this: At Re = 100, the numerical and theoretical curves match well. Using a lower detection threshold (e.g. 95%) or referencing the analytical peak velocity (1.5U) instead of the outlet value would give a more meaningful numerical δ at high Re.



5.5 Residual vs Iteration — Re = 1000

The max vorticity residual starts near 10^2 and drops rapidly in the first 200 iterations as the initial transient is resolved. A small plateau and slight bump appear around iteration 500–600, where the interior vorticity field reorganises as wall vorticity diffuses inward. After that, the residual declines smoothly and reaches the convergence criterion (10^{-6}) at approximately iteration 1900. The smooth monotonic decay confirms numerical stability and correct implementation of the solver.

What changes this: Higher Re requires more iterations to converge because convection slows down the diffusive spreading of vorticity. A finer grid also increases iteration count. Tightening tol to 10^{-8} would extend the residual curve further. Using a multigrid or SOR accelerator would reduce the iteration count significantly.



6. Conclusion

The ψ-Ω CFD solver successfully captures the developing laminar flow between parallel plates. Key conclusions:

- Entry length scales linearly with Re: Le = 0.05·Re·2h. Doubling Re doubles the required channel length.
- At Re = 100, the flow fully develops within L = 10 and matches the Poiseuille analytical solution.
- At Re = 500 and 1000, the domain is too short; the flow remains in the entry region and the outlet profile is far from parabolic.
- The velocity profiles shift rightward (centreline accelerates more) as Re decreases, confirming faster development at lower Re.
- The residual convergence confirms a stable, monotonically converging steady-state solution, with more iterations needed at higher Re.
- Grid resolution in the y-direction is the critical parameter for accuracy at high Re; near-wall grid clustering is recommended for Re > 500.